S. B. JAIN INSTITUTE OF TECHNOLOGY, MANAGEMENT & RESEARCH, NAGPUR.

Practical No. 4

AIM: An online store is launching a new product collection that should provide a seamless viewing experience on both smartphones and desktop computers. Create a responsive Product Showcase webpage using Tailwind CSS Flexbox and responsive breakpoints.

NAME OF STUDENT	
USN NO.	
DATE OF PERFORMANCE	
DATE OF SUBMISSION	

OBSERVATION (Practical No. 4)

Application Software:	
System Software (OS):	
Web Application URL:	

OBSERVATIONS:

Regular Performance in Lab (3)	
Faculty Signature & Date	

AIM: An online store is launching a new product collection that should provide a seamless viewing experience on both smartphones and desktop computers. Create a responsive Product Showcase webpage using Tailwind CSS Flexbox and responsive breakpoints.

OBJECTIVES

- To master the implementation of responsive layout systems using Tailwind CSS media query breakpoints (sm:, md:, lg:).
- To utilize Flexbox properties for structural axis alignment, dynamic component distribution, and cross-axis positioning.
- To create a modern web dashboard element that reflows automatically between compact mobile screens and wide desktop displays.

THEORY

1. Responsive Web Infrastructure & Tailwind Breakpoints

Responsive web design centers on building fluid interfaces that read and adapt cleanly to any screen size or physical device form factor. Tailwind CSS handles responsiveness using mobile-first media query breakpoints. This design pattern applies global styles to mobile layout architectures by default, and overrides them for broader screens by appending specific viewport modifiers:

- sm: (Targets minimum screen width $\ge$ 640px)
- md: (Targets minimum screen width $\ge$ 768px)
- lg: (Targets minimum screen width $\ge$ 1024px)
- xl: (Targets minimum screen width $\ge$ 1280px)

By defining layouts mobile-first, the client's browser avoids parsing unnecessary design overrides on resource-constrained mobile hardware, resulting in faster rendering loops and improved performance.

2. Tailwind Flexbox Mechanics

The CSS Flexible Box Layout Engine (Flexbox) provides a predictable way to align items along a single layout dimension (either a horizontal row or a vertical column). Tailwind maps these properties to highly reusable structural utility utilities:

- flex: Establishes a flex container context, changing child nodes into immediate flex items.
- flex-col / flex-row: Modifies the main architectural axis vector, defining whether cards flow vertically down or horizontally across.

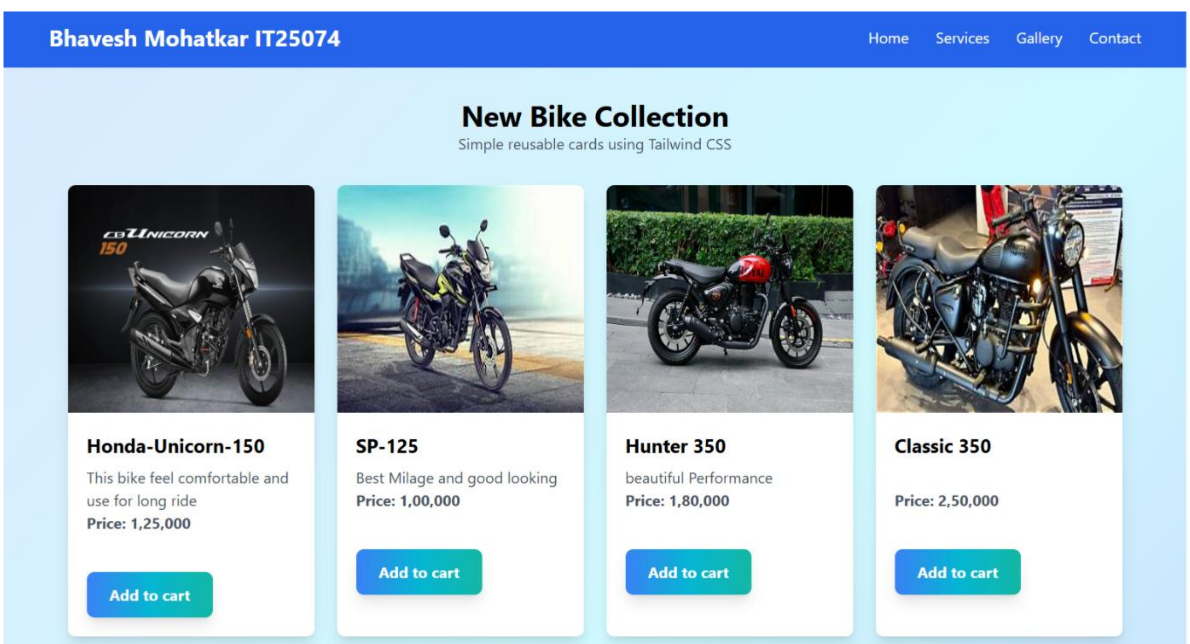
- items-center / items-start: Governs alignment along the cross-axis (perpendicular to the main axis).
- justify-between / justify-center: Distributes leftover free space across the active operational main axis.
- gap-x / gap-y: Assigns uniform spatial padding between adjacent items without disrupting border metrics.

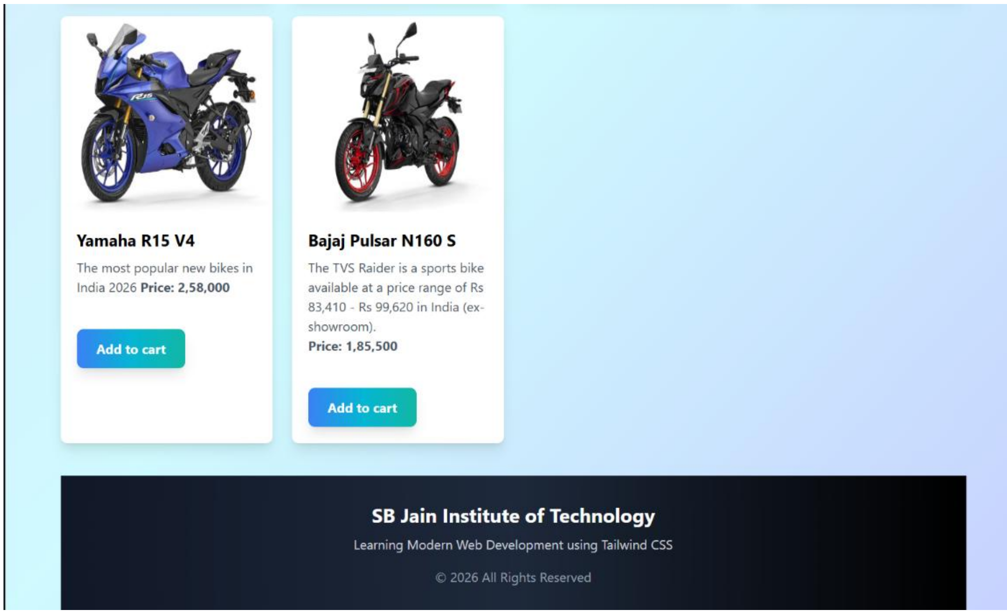
SOURCE CODE GITHUB REPOSITORY

https://github.com/bhaveshmohatkar66-creator/Bhaveshtrial

SOURCE CODE GITHUB REPOSITORY QR CODE

OUTPUT SCREENSHOTS





REVIEW QUESTIONS

1. How do Tailwind CSS responsive breakpoints like md: and lg: modify the default mobile-first CSS behavior?

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2. What is the explicit operational difference between the flex-row and flex-col axis utility classes?

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3. How does the flex-wrap utility parameter prevent layout structural breakage when styling rows of dynamic interface buttons?

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4. Why are flex layouts preferred over static sizing structures when building multi-screen commercial storefronts?

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5. How does the active scaling utility configuration (active:scale-95) improve perceived user experience during button interactions?

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CONCLUSION

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REFERENCES

- https://tailwindcss.com/docs/responsive-design
- https://tailwindcss.com/docs/flexbox-direction
- https://developer.mozilla.org/en-US/docs/Web/CSS/CSS_Flexible_Box_Layout/Basic_Concepts_of_Flexbox

EVALUATION

Regular Performance in Lab (3)	
Viva-Voce (3)	
Quality of Submission (4)	
Faculty Signature & Date	